

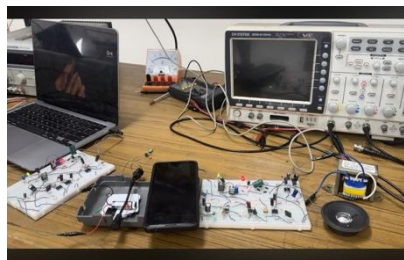
EEE 310 (September 2026)

Communication System I Laboratory

Final Project Report

Section: B2 Group: 05

Li-Fi: Audio Transmission over a Laser Beam using Analog Intensity Modulation



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Date of Submission: 28 September 2026

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Academic Honesty Statement

We, the undersigned members of Group 05, Section B2, declare that the work presented in this report is our own. The circuit design, simulation, hardware implementation, measurements and analysis reported here were carried out by our group. All external sources, datasheets and reference designs that informed our work have been cited in the References section. We have not copied any part of this report from another group or from any publication without acknowledgement. And AI tools have been utilized while arranging this Report.

We understand that the design of this project involved a laser source, and we confirm that the laser was operated responsibly throughout the project, with the beam kept away from the eyes of all persons present.

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Table 1: Declaration of academic honesty by all group members.

Table of Contents

1	Abstract	4
2	Introduction	4
3	Design.....	6
3.1	Design Method (PO(a)).....	6
3.1.1	Linear intensity modulation of the laser diode.....	6
3.1.2	Preamplifier and input level setting	6
3.1.3	Band-limiting filters	7
3.1.4	Photodiode front end and receiver gain.....	7
3.1.5	Output stage and supply arrangement	8
3.2	Circuit Diagram.....	8
3.3	Simulation Model.....	10
3.4	Hardware Design.....	12
4	Implementation.....	13
4.1	Description	13
5	Design Analysis and Evaluation	15
5.1	Novelty	15
5.2	Design Considerations (PO(c))	15
5.2.1	Considerations to public health and safety.....	15
5.2.2	Considerations to environment.....	16
5.2.3	Considerations to cultural and societal needs	16
5.3	Investigations (PO(d)).....	17
5.3.1	Design of Experiment.....	17
5.3.2	Data Collection.....	17
5.3.3	Results and Analysis	22
5.3.4	Interpretation and Conclusions on Data	23
5.4	Impact Assessment (PO(f))	24
5.4.1	Assessment of Societal and Cultural Issues	24
5.4.2	Assessment of Health and Safety Issues	24
5.4.3	Assessment of Legal Issues.....	25
5.5	Sustainability Evaluation (PO(g))	25
6	Reflection on Individual and Team Work (PO(i))	26
6.1	Individual Contribution of Each Member	26

6.2 Mode of Team Work.....	26
6.3 Diversity Statement of Team.....	26
6.4 Log Book of Project Implementation.....	27
7 Communication to External Stakeholders (PO(j))	28
7.1 Executive Summary	28
7.2 User Manual	28
7.3 GitHub Link	29
7.4 YouTube Link	29
8 Project Management and Cost Analysis (PO(k))	29
8.1 Bill of Materials	29
8.2 Calculation of Per Unit Cost of Prototype	30
8.3 Calculation of Per Unit Cost of Mass-Produced Unit	30
8.4 Timeline of Project Implementation	31
9 References	32
Appendix A Mapping of Course Outcomes and Program Outcomes	33

1 Abstract

This project presents the design, simulation and hardware implementation of a Li-Fi (Light Fidelity) audio communication link in which an analog audio signal is transmitted over a free-space laser beam and recovered as audible sound at the receiver. In the transmitter, the audio signal from a mobile phone, laptop or electret microphone is attenuated to a fixed level, amplified by a TL072 operational amplifier, band-limited by a second-order Sallen-Key low-pass filter, and then used to modulate the current of a 650 nm, 5 mW laser diode through a closed-loop constant-current driver built around an LM358 and a BD135 transistor. Because the laser is driven by current rather than voltage, the optical power varies linearly with the message signal, which removes the severe distortion that a simple voltage-driven laser produces. At the receiver, a reverse-biased BPW34 photodiode converts the modulated light back into a photocurrent, which is amplified by a band-pass stage, filtered by a second Sallen-Key low-pass filter with a cut-off of 10.7 kHz, and delivered to an LM386 power amplifier driving an 8 Ω loudspeaker. The complete system was simulated in PSpice and then built on breadboards. The measured link reproduces speech and music intelligibly over a distance of about 10 metres indoors, with a measured -3 dB audio bandwidth of approximately 40 Hz to 12 kHz. The project demonstrates that a fully analog, low-cost optical communication link can be realised with commonly available components, and it illustrates the practical engineering trade-offs among linearity, bandwidth, ambient-light immunity and signal-to-noise ratio.

2 Introduction

Radio-frequency spectrum congestion is one of the central problems of modern wireless communication. As the number of connected devices grows, the licensed radio bands become increasingly crowded, and in certain environments — hospitals, aircraft cabins, petrochemical plants and secure facilities — radio transmission is either restricted or undesirable. Li-Fi, or Light Fidelity, addresses this problem by using the visible-light portion of the electromagnetic spectrum as the communication channel. Because light does not penetrate opaque walls, a light-based link is inherently confined to the room in which it operates, which provides both spatial reuse of the channel and a degree of physical-layer security that radio cannot offer.

The objective of this project is to design and build a complete analog Li-Fi link that transmits audio over a laser beam. The transmitter must convert an audio voltage into a proportional variation of optical power; the receiver must convert the received optical power back into an audio voltage and drive a loudspeaker with acceptable fidelity. The link is required to operate from a low-voltage DC supply, to cover the useful speech and music band, and to work at a usable distance across a laboratory or classroom.

This is a complex engineering problem in the sense of the Washington Accord attributes. It requires depth of knowledge (K3–K5): semiconductor device physics governs the threshold behaviour of the laser diode and the responsivity of the photodiode; analog circuit theory governs the feedback loop of the current driver and the transfer functions of the active filters; and communication theory governs the choice of bandwidth and the treatment of noise. It involves conflicting requirements: a large modulation depth improves the signal-to-noise ratio but pushes the laser below threshold and clips the waveform, and a small photodiode load resistance preserves bandwidth but reduces the recovered signal amplitude. It has no obvious solution, since the first working prototype produced badly distorted audio and the cause — voltage-mode drive of a non-linear device — was not apparent from the schematic alone. Finally, the problem involves diverse stakeholders: users of the link, bystanders exposed to the laser beam, and the institution that must dispose of the electronics at end of life.

Several alternative solutions were considered before the present design was adopted:

- Bluetooth or Wi-Fi audio, which is mature and robust, but uses the congested radio spectrum and would not demonstrate optical communication.
- Infrared LED transmission, which is safer for the eyes and cheaper, but has a wide beam, low irradiance at the receiver, and therefore a much shorter working distance.
- Optical fibre transmission, which is immune to ambient light and offers enormous bandwidth, but requires a guided medium and does not demonstrate free-space propagation.
- Digital modulation over the optical channel, for example pulse-width or pulse-code modulation, which is more robust against ambient light but demands high switching bandwidth, additional encoding and decoding hardware, and would have obscured the analog design content that this course emphasises.

Direct analog intensity modulation of a laser diode was selected because it demonstrates the complete communication chain — source, modulator, channel, detector and demodulator — with the smallest component count, while still exposing the linearity and noise problems that make optical communication interesting.

3 Design

3.1 Design Method (PO(a))

The design applies knowledge of semiconductor physics, analog electronics and communication theory in five stages. Each stage is derived below.

3.1.1 Linear intensity modulation of the laser diode

A laser diode is a strongly non-linear device. Below its threshold current I_{th} the device emits only weak spontaneous (LED-like) light; above threshold the optical output power rises almost linearly with the drive current:

$$P_{opt} \approx \eta (I - I_{th}) \text{ for } I > I_{th}$$

where η is the slope efficiency in watts per ampere. Two consequences follow directly. First, the laser must be biased at a quiescent current I_{bias} that lies above threshold, so that the negative half-cycles of the audio signal never drive the device into the dark region. Second, the modulation must be applied as a current, not as a voltage, because the exponential current-voltage characteristic of the junction would otherwise convert a sinusoidal voltage into a strongly distorted current.

The transmitter therefore uses a closed-loop transconductance (voltage-to-current) stage. The op-amp U2 forces its inverting input to follow its non-inverting input, so the voltage developed across the emitter resistor R11 equals the input voltage, and the laser current becomes

$$I_{laser} \approx V_{in} / R11$$

which is independent of the laser's own forward characteristic, of temperature, and of the transistor current gain. With the bias divider R10 = 68 k Ω and R9 = 10 k Ω from a 9 V rail, the quiescent voltage at the non-inverting input is

$$V_{bias} = 9 \times 10 / (68 + 10) = 1.154 \text{ V}$$

and with R11 = 38 Ω the intended quiescent laser current is $1.154 / 38 \approx 30 \text{ mA}$, which lies above the threshold of the module used and below its 40 mA maximum rating. The audio signal, superimposed on this bias through the coupling capacitor C3, then swings the current symmetrically about the bias point.

3.1.2 Preamplifier and input level setting

The audio source is a mobile phone or laptop headphone output, which was measured at 0.5 V RMS at full volume. The signal is first attenuated by a fixed resistive divider and then amplified by U1A in the non-inverting configuration, whose gain is

$$A_v = 1 + R6/R1 = 1 + 2.2 \text{ k}\Omega / 1 \text{ k}\Omega = 3.2$$

A fixed divider was chosen in place of a potentiometer so that the modulation depth cannot be increased accidentally beyond the safe current rating of the laser. $R_{13} = 100 \text{ k}\Omega$ sets the input bias to the mid-rail reference of 4.5 V, and $C_6 = 47 \text{ pF}$ across the feedback resistor limits the closed-loop bandwidth and prevents high-frequency instability. Because the circuit runs from a single 9 V supply, a 4.5 V reference is generated by a 1:1 resistive divider decoupled by $C_5 = 47 \text{ }\mu\text{F}$, and every signal node is referred to this virtual ground.

3.1.3 Band-limiting filters

Two unity-gain Sallen-Key low-pass filters are used, one in each unit. For equal resistors R and capacitors C_1 (feedback) and C_2 (to ground), the cut-off frequency and quality factor are

$$f_c = 1 / (2\pi R \sqrt{C_1 C_2}) \quad Q = \frac{1}{2} \sqrt{C_1 / C_2}$$

In the transmitter, $R_7 = R_8 = 10 \text{ k}\Omega$, $C_7 = 1 \text{ nF}$ and $C_8 = 470 \text{ pF}$ give $f_c \approx 23 \text{ kHz}$ with $Q = 0.73$, which is close to the maximally flat Butterworth value of 0.707. This stage keeps wideband noise generated in the preamplifier away from the laser without encroaching on the audio band. In the receiver, $R_{23} = R_{24} = 10 \text{ k}\Omega$, $C_{15} = 2.2 \text{ nF}$ and $C_{16} = 1 \text{ nF}$ give

$$f_c = 1 / (2\pi \times 10 \text{ k}\Omega \times \sqrt{2.2 \text{ nF} \times 1 \text{ nF}}) \approx 10.7 \text{ kHz}, \quad Q = 0.74$$

so the receiver filter dominates the overall high-frequency response. Placing the sharper filter after the photodiode is deliberate: noise picked up in the optical channel and generated in the detector itself can only be removed at the receiving end. The two cut-off frequencies are deliberately staggered rather than made equal, because two identical second-order filters in cascade would be 6 dB down at the nominal cut-off and would reduce the useful bandwidth of the link.

3.1.4 Photodiode front end and receiver gain

The BPW34 photodiode is operated in reverse bias (photoconductive mode), in which the junction capacitance is small and the response is fast and linear. The photocurrent is converted to a voltage across the load resistor R_{18} , and the pole formed by this resistor and the diode capacitance $C_j \approx 25 \text{ pF}$ sets an upper limit on the detector bandwidth:

$$f_{\text{detector}} = 1 / (2\pi R_{18} C_j) = 1 / (2\pi \times 470 \text{ }\Omega \times 25 \text{ pF}) \approx 13.5 \text{ kHz}$$

$R_{18} = 470 \text{ }\Omega$ was selected as a compromise: a larger resistor would give a larger signal voltage but would move this pole into the audio band and dull the sound, while a smaller resistor would waste signal. The recovered voltage is then amplified by U_{3A} in the non-inverting configuration with

$$A_v = 1 + R_{16}/R_{15} = 1 + 47 \text{ k}\Omega / 2.2 \text{ k}\Omega \approx 22$$

The gain-setting network is AC-coupled by $C_{13} = 470 \text{ nF}$, which introduces a high-pass corner at

$$f_{\text{HP}} = 1 / (2\pi \times 2.2 \text{ k}\Omega \times 470 \text{ nF}) \approx 154 \text{ Hz}$$

This high-pass action is not incidental — it is the principal defence against ambient light. Mains-powered lamps flicker at 100 Hz, and daylight appears as a slowly varying DC term; both fall below this corner and are therefore attenuated before they can saturate the following stages. $C10 = 100 \text{ pF}$ across $R16$ limits the closed-loop bandwidth of the stage and ensures stability.

3.1.5 Output stage and supply arrangement

The filtered audio is passed through a $10 \text{ k}\Omega$ volume control to an LM386 power amplifier operating at its default gain of 20 with pins 1 and 8 open, driving an $8 \text{ }\Omega$ loudspeaker through a $220 \text{ }\mu\text{F}$ coupling capacitor. The receiver front end is powered through a $100 \text{ }\Omega$ series resistor decoupled by a $470 \text{ }\mu\text{F}$ capacitor, so that the large current pulses drawn by the loudspeaker amplifier cannot modulate the supply of the sensitive photodiode stage. The transmitter and receiver are powered separately so that no common ground path exists between them, which removes a well-known source of hum in audio links.

3.2 Circuit Diagram

Figure 1 shows the block diagram of the complete system, and Figure 2 shows the full circuit schematic as entered in PSpice. In the schematic the optical channel is modelled by a current-controlled current source F1, which senses the laser current and injects a proportional photocurrent into the receiver node; this allows the entire link, from audio input to loudspeaker, to be simulated as a single circuit.

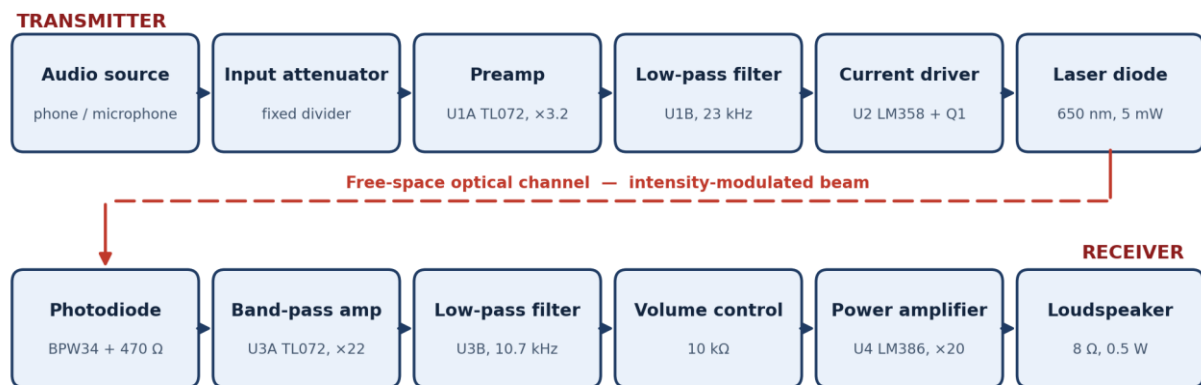


Figure 1: Block diagram of the Li-Fi audio transmission system.

3.3 Simulation Model

The design was verified in PSpice in three analyses before any component was soldered.

Bias point analysis (Figure 4) confirms that the operating points are correct. The mid-rail reference sits at 4.500 V, every op-amp output rests near that value, the non-inverting input of the driver is at 1.154 V and the emitter of Q1 at 1.155 V, which shows that the current feedback loop is closing correctly. The corresponding laser current is $1.155 \text{ V} / 38 \text{ } \Omega \approx 30 \text{ mA}$, and the collector-emitter voltage of Q1 remains high enough for the transistor to stay in the active region.

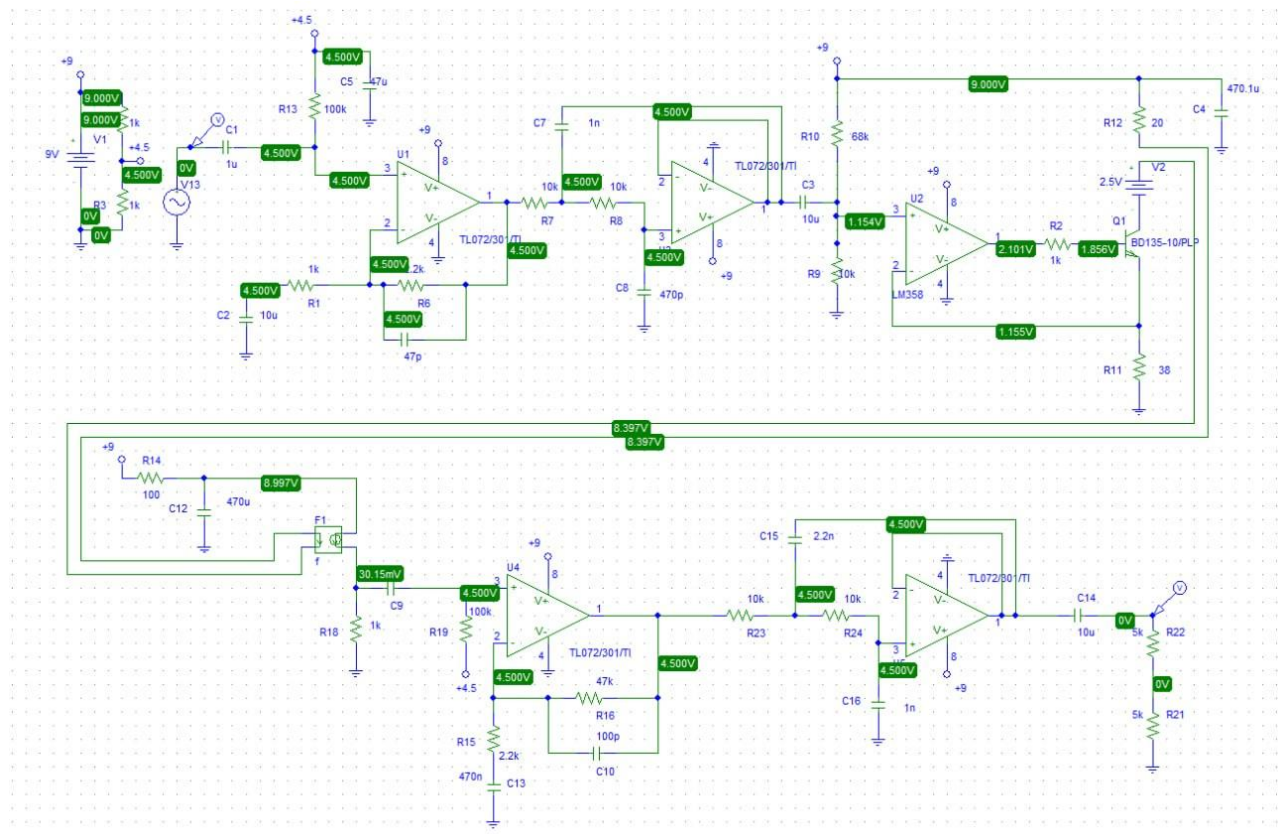


Figure 4: PSpice bias point analysis of the complete link, showing node voltages.

AC sweep analysis (Figure 5) gives the small-signal frequency response of the entire link from the audio input to the loudspeaker. The response is flat through the mid-band and falls away below a few hundred hertz owing to the AC-coupling networks and above approximately 10 kHz owing to the receiver filter. Figure 6 shows the same response plotted on a linear frequency axis, which makes the steepness of the high-frequency roll-off clear: the output has fallen to a negligible value by 30 kHz.

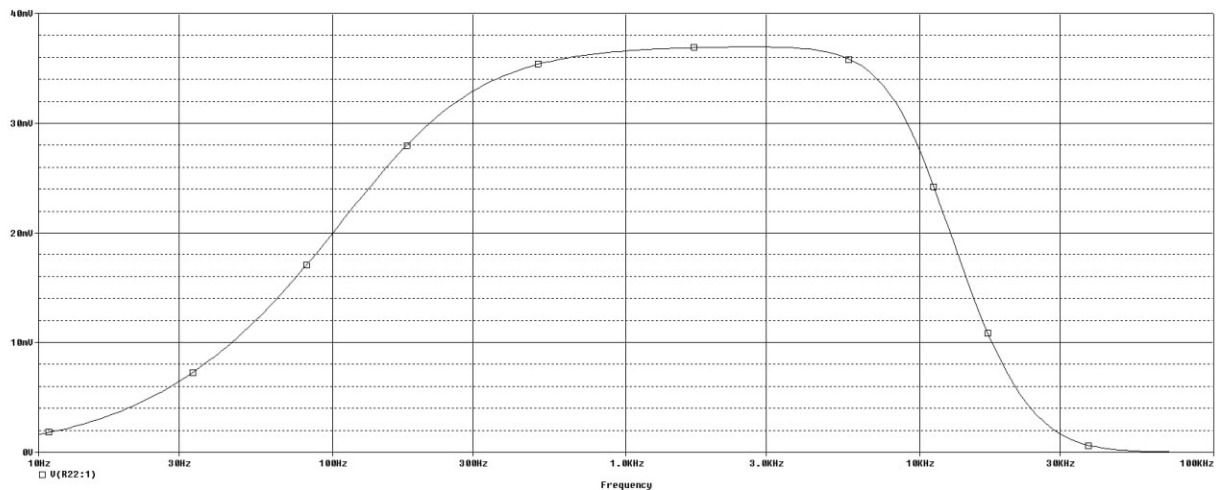


Figure 5: Simulated frequency response of the complete link (logarithmic frequency axis).

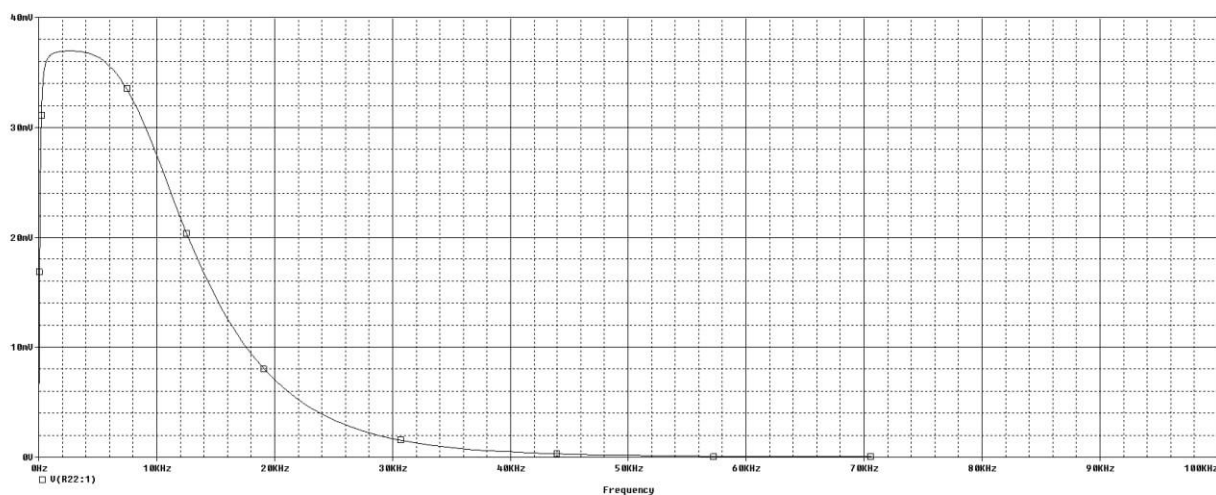


Figure 6: Simulated frequency response on a linear frequency axis, showing the roll-off above the audio band.

Transient analysis (Figure 7) compares the input audio waveform with the recovered waveform at the loudspeaker for a 500 Hz sinusoid. The recovered waveform is a clean sinusoid of the same frequency with no visible flattening of either peak, which confirms that the laser remains above threshold throughout the cycle and that no stage is clipping.

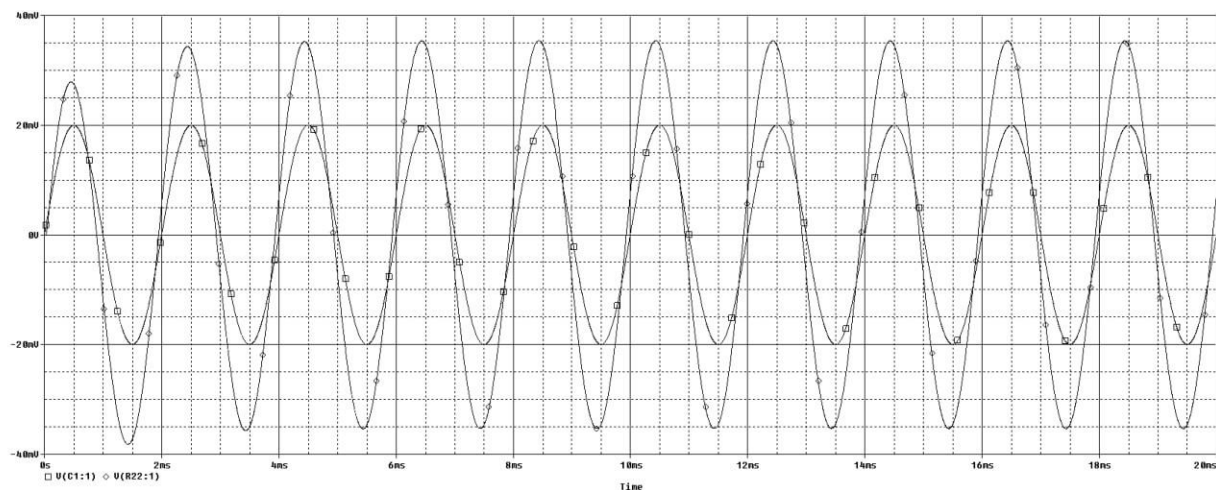


Figure 7: Simulated transient response: input signal and recovered output at the loudspeaker.

3.4 Hardware Design

The system was built as two independent breadboard assemblies, each with its own 9 V supply, its own ground and its own on-board decoupling. Figure 8 shows the transmitter, with the laser module mounted at the board edge and a 3.5 mm socket for the audio source. Figure 9 shows the receiver. Figure 10 shows the laser module and Figure 11 the electret microphone that provides the alternative live-audio input.

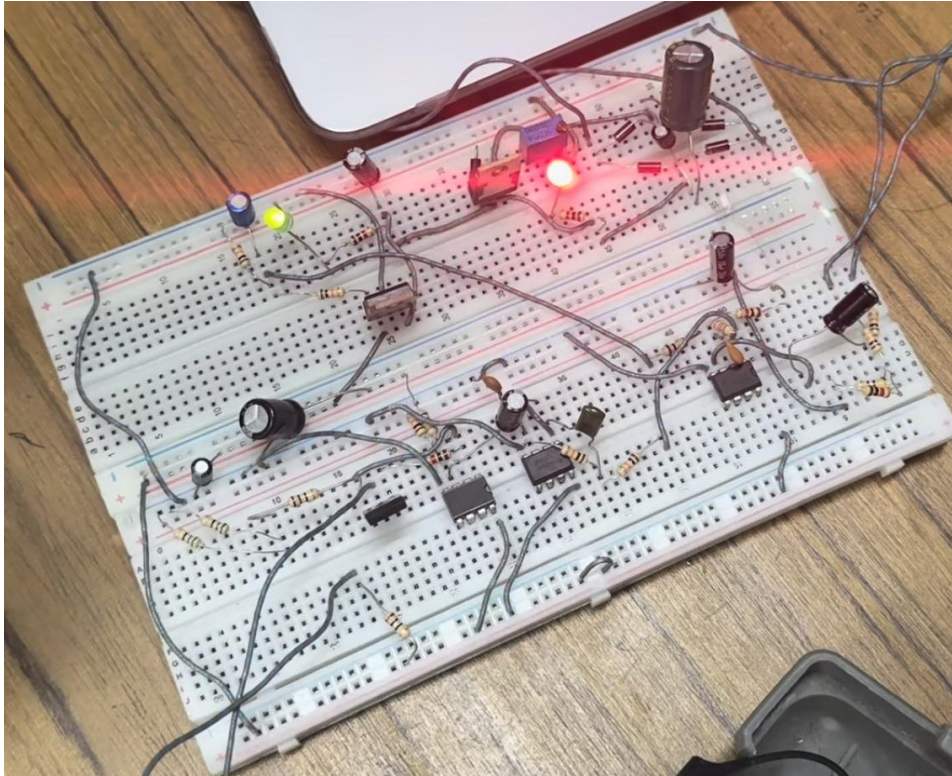


Figure 8: Transmitter board in operation; the red glow at the centre is the laser module under modulation.

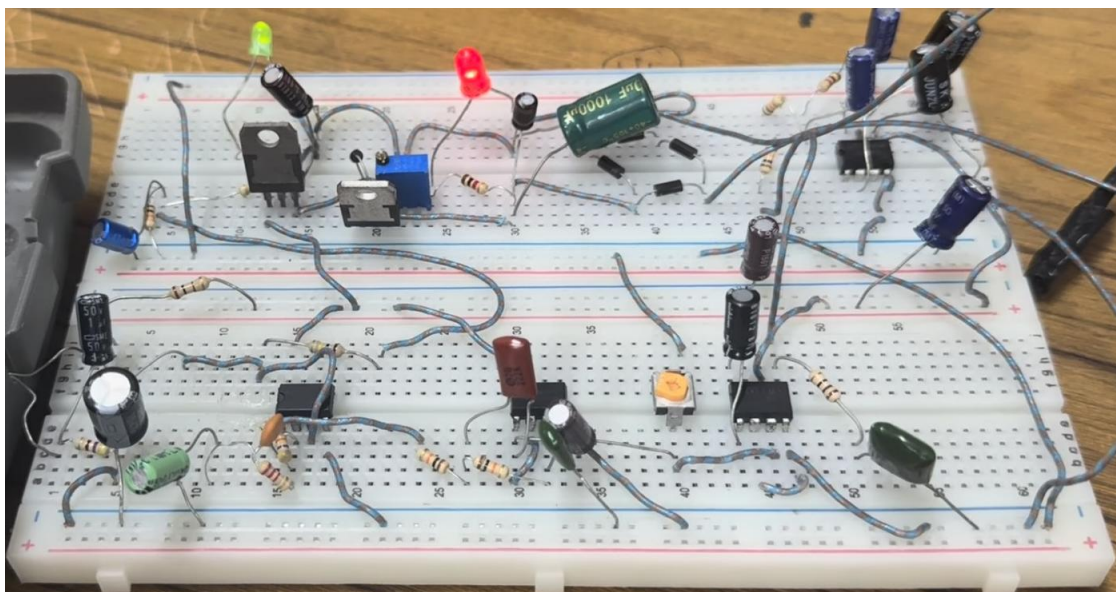


Figure 9: Receiver board, including the photodiode front end, filter, LM386 output stage and on-board regulator.



Figure 10: The 650 nm, 5 mW laser diode module used as the optical source.



Figure 11: Electret microphone used as the alternative live audio input to the transmitter.

4 Implementation

4.1 Description

The completed link is shown in Figure 12. The audio source on the left — a laptop, a mobile phone or the microphone — feeds the transmitter board. The laser module is aimed across the bench at the photodiode mounted on the receiver board, and the recovered audio is reproduced by the loudspeaker on the right. Measurement instruments are connected at the nodes of interest during testing.

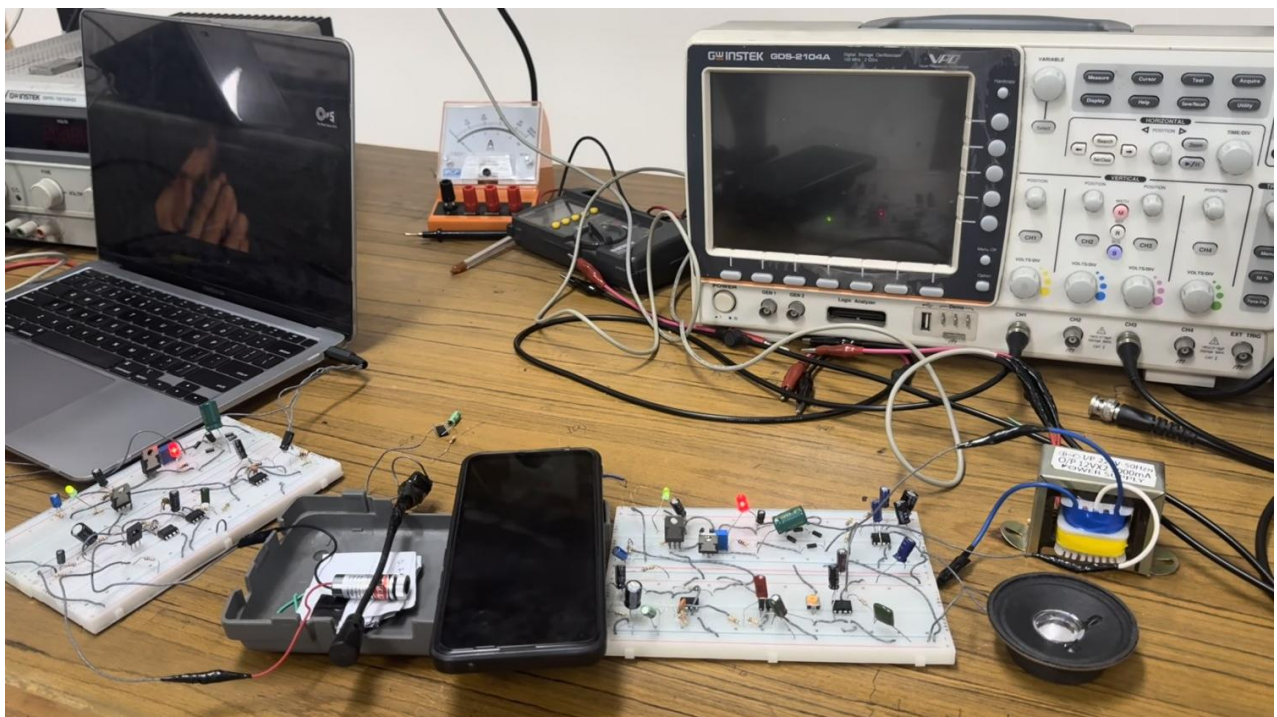


Figure 12: Complete experimental setup: audio source, transmitter, optical path, receiver, loudspeaker and instrumentation.

Figure 13 shows the link operating with the beam clearly visible, and Figure 14 shows the beam landing on the active area of the photodiode. Correct alignment of this spot proved to be one of

the most important practical factors in the performance of the link: moving the spot off the centre of the die reduces the received signal sharply and increases the audible noise.

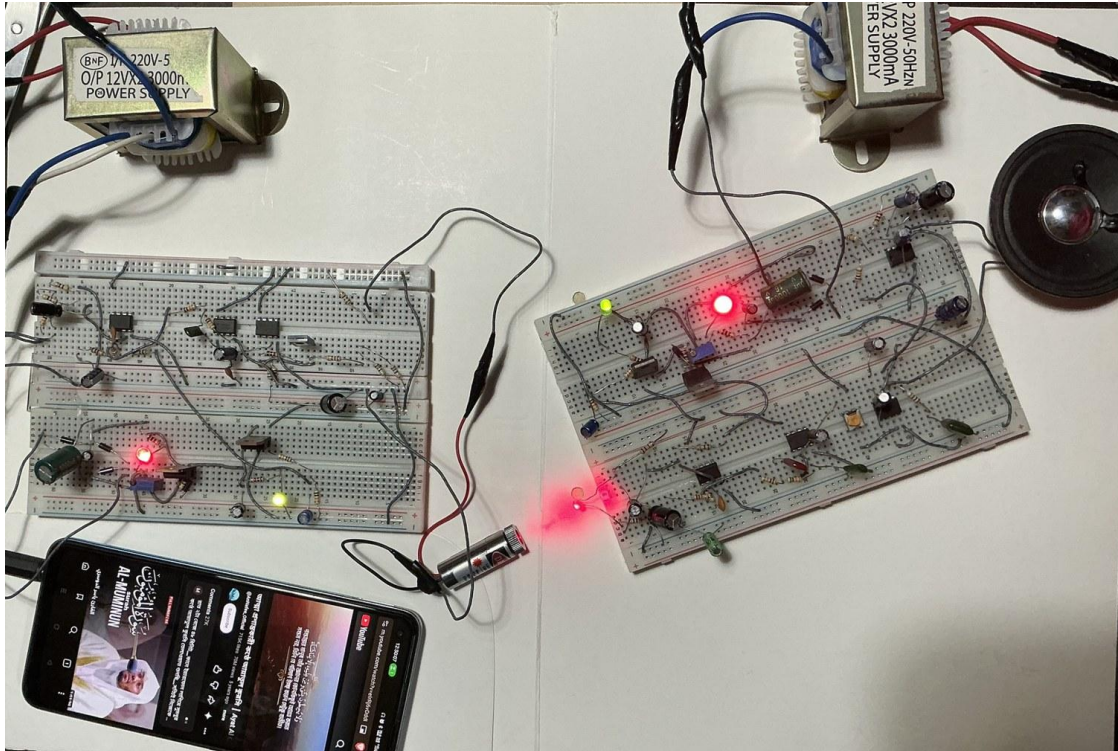


Figure 13: The link in operation with the laser beam visible between transmitter and receiver.



Figure 14: Close-up of the laser spot falling on the active area of the BPW34 photodiode.

The practical construction sequence was as follows. The constant-current laser driver was built and verified first, on its own, by confirming the voltage across R11 with a multimeter before any signal was applied. The preamplifier and filter were then added and the transmitter checked for a clean, unclipped waveform at the output of U1B. The receiver was built next and aligned by adjusting the photodiode position until the DC voltage at the photodiode node sat in the middle of its range. Only then were the two halves brought together and the audio quality assessed.

5 Design Analysis and Evaluation

5.1 Novelty

The novelty of this project lies in the treatment of the optical source as a current-controlled device inside a feedback loop, rather than as a lamp switched by a voltage. Most student-level Li-Fi demonstrations drive an LED or laser module directly from an audio amplifier output, which produces audible distortion because the optical source is non-linear and switches off completely during negative half-cycles. In the present design the laser current is servo-controlled by an operational amplifier, so the optical power follows the message signal linearly over the whole cycle, and the modulation depth is set by a resistor rather than by a volume control that can be turned too far.

Three further design decisions distinguish this implementation:

- The filters are staggered between the two units — a wider filter at the transmitter and the dominant filter at the receiver — rather than duplicated, so that noise is removed where it is generated without narrowing the useful bandwidth.
- The receiver front end deliberately high-passes the signal at about 154 Hz, which rejects the 100 Hz flicker of mains-powered lighting; ambient-light immunity is obtained by filtering rather than by shielding alone.
- The transmitter input level is fixed by a resistive divider instead of a potentiometer, which makes the modulation depth — and therefore the safety of the laser diode — independent of the user.

The system is fully analog. It contains no microcontroller and no firmware, so its behaviour can be predicted entirely from circuit theory and verified by simulation, which makes it valuable as a teaching artefact as well as a working link.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

The dominant safety consideration in this project is the laser source. A 5 mW, 650 nm laser is a Class 3R device, and direct viewing of the beam or of a specular reflection can injure the retina. The following measures were adopted throughout the work and are stated in the user manual:

- The optical axis was kept at bench height, well below standing eye level, and never directed towards any person.
- The beam path was terminated at the receiver photodiode so that the beam does not continue into the room.

- Reflective objects — mirrors, polished metal, phone screens — were kept out of the beam path.
- The constant-current driver limits the laser current to below the 40 mA rating of the module under all signal conditions, so the device cannot be driven to an unsafe optical power by turning up the volume.

Electrically, both units operate from 9 V DC, so no hazardous voltage is exposed on the boards. The bench power supply built for the project does connect to the 230 V mains through a step-down transformer; its primary side was insulated, kept away from the working area, and switched off when connections were being changed. No component was found to overheat in extended operation: the BD135 transistor dissipates well under 100 mW and remained at ambient temperature throughout.

5.2.2 Considerations to environment

The prototype was built on reusable breadboards with through-hole components, so no etching chemicals were consumed and no permanent electronic waste was created; every component can be recovered and used again. Power consumption is very low — the receiver draws 10.48 mA at 7.96 V, about 83 mW, and the transmitter operates from a similar supply — so the energy footprint of the link is negligible compared with a radio transceiver of equivalent function. The laser module is rated for more than 2000 hours of operation, which limits the replacement rate. The only materials of environmental concern are the electrolytic capacitors and the semiconductor devices, which contain small amounts of recoverable metal and should be routed to electronic-waste collection rather than to general refuse at end of life.

5.2.3 Considerations to cultural and societal needs

The entire system is built from components that are inexpensive and available in local electronics markets in Dhaka; no specialised optical assembly, integrated transceiver or imported module is required. This matters for a country in which laboratory budgets are limited: a working demonstration of optical wireless communication can be produced for a few hundred taka and reproduced by any undergraduate group. The design also addresses a genuine societal need. Light-based communication is useful precisely where radio is restricted or unavailable — in hospital wards near sensitive instruments, in industrial areas where radio transmitters are prohibited, and in classrooms or offices where a link that cannot pass through a wall is a security advantage rather than a limitation.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

Four experiments were designed to characterise the link and to compare its measured behaviour with the simulated prediction:

- **Signal-chain tracing.** A sinusoidal test tone was applied at the transmitter input and the waveform observed at four successive points — the preamplifier input, the filter output, the emitter resistor R11 (which represents the laser current) and the photodiode node — to verify that each stage performs its intended function without clipping.
- **Frequency response.** A sinusoid of constant amplitude was swept in frequency from 40 Hz to 13 kHz and the amplitude of the recovered signal at the loudspeaker recorded, in order to locate the -3 dB points of the complete link.
- **Ambient-light sensitivity.** The recovered waveform was observed with the room lights dimmed and again with a lamp switched on close to the receiver, to quantify the effect of artificial illumination on the noise floor.
- **Detector comparison and range.** Three detector arrangements were compared — a phototransistor, a bare photodiode and a photodiode inside a black shroud — and the maximum usable range of the final arrangement was determined by listening tests.

The instrument used throughout was a GW Instek GDS-2104A 100 MHz digital storage oscilloscope, supplemented by a digital multimeter for DC measurements. The audio source was a MacBook headphone output, measured at 0.5 V RMS at maximum volume.

5.3.2 Data Collection

Table 2 summarises the DC operating conditions measured on the completed hardware.

Quantity	Measured value	Instrument
Transmitter supply voltage	8.83 V	Multimeter
Receiver supply voltage	7.96 V	Multimeter
Receiver supply current	10.48 mA	Multimeter
Laser quiescent (bias) current	7.75 mA	V across R11 $\div$ 38 Ω
Laser peak current with signal	≈ 10 mA	V across R11 $\div$ 38 Ω
Laser forward voltage	2.06 V	Multimeter

Table 2: Measured DC operating conditions of the completed link.

Figures 15 to 18 show the signal at four points along the chain for a mid-band test tone. The waveform at the input of U1A is the raw audio; at the output of U1B it is amplified and band-limited; across R11 it represents the laser drive current; and at the photodiode node it is the recovered photocurrent signal before amplification.

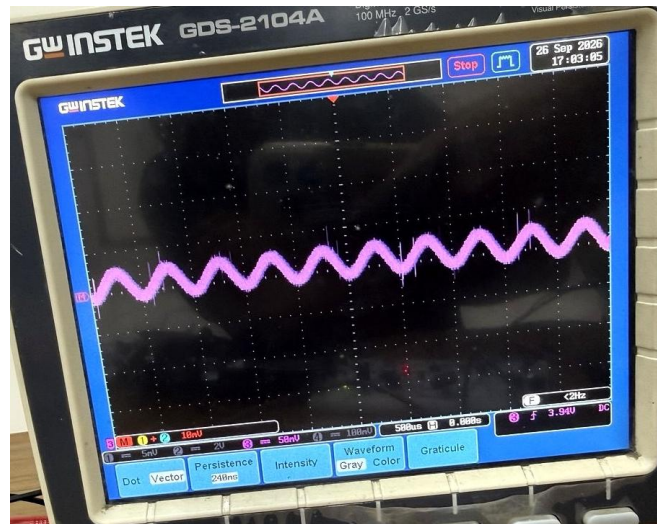


Figure 15: Audio signal at the preamplifier input (U1A pin 3).

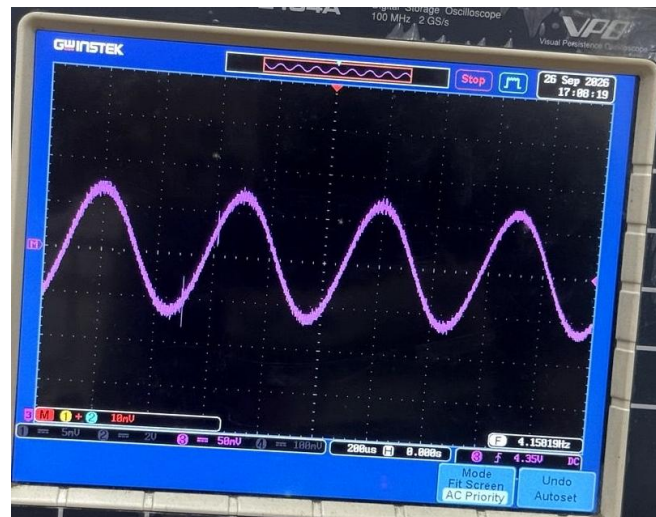


Figure 16: Filtered audio at the output of the transmitter filter (U1B pin 7).

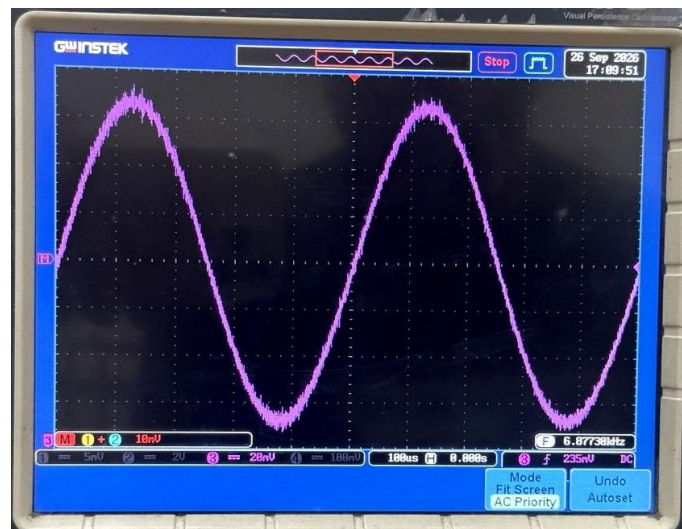


Figure 17: Voltage across R11, which is proportional to the laser drive current.

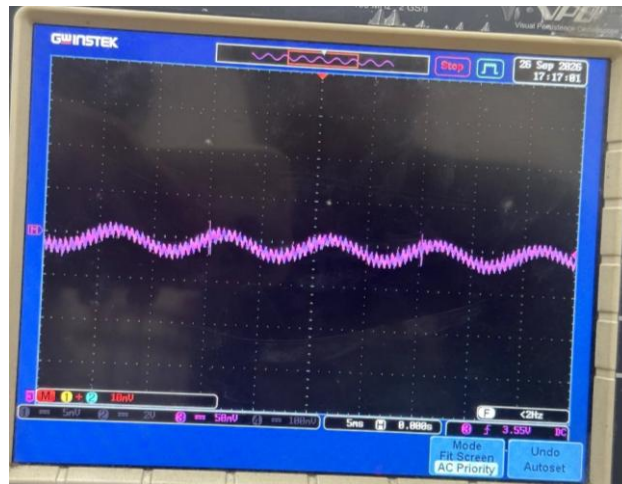


Figure 18: Recovered signal at the photodiode node of the receiver.

Figure 19 shows the transmitted and recovered waveforms captured on the same screen. The recovered waveform preserves the shape of the original; the small phase shift between them is the accumulated delay of the two filter stages.

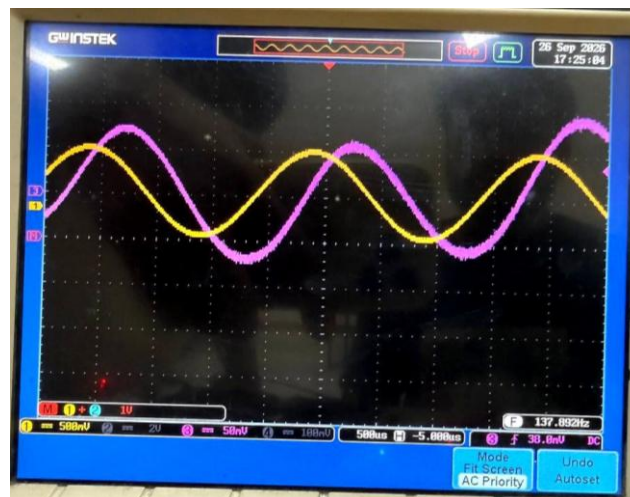


Figure 19: Transmitted signal (yellow) and recovered signal (magenta) displayed together.

Figures 20 to 25 show the recovered output at frequencies from 40 Hz to 13 kHz, from which the band edges of the link were determined.

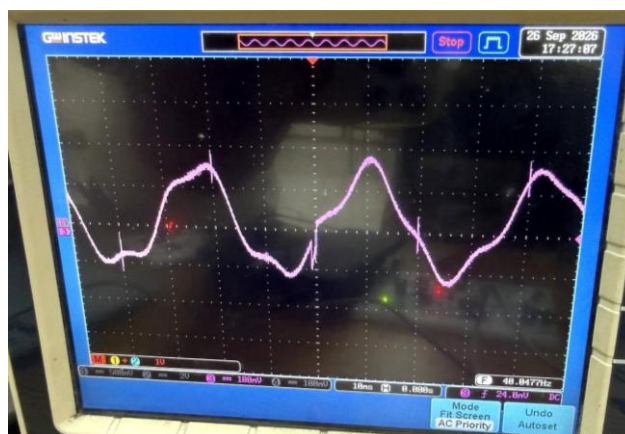


Figure 20: Recovered output at 40 Hz — the lower band edge.

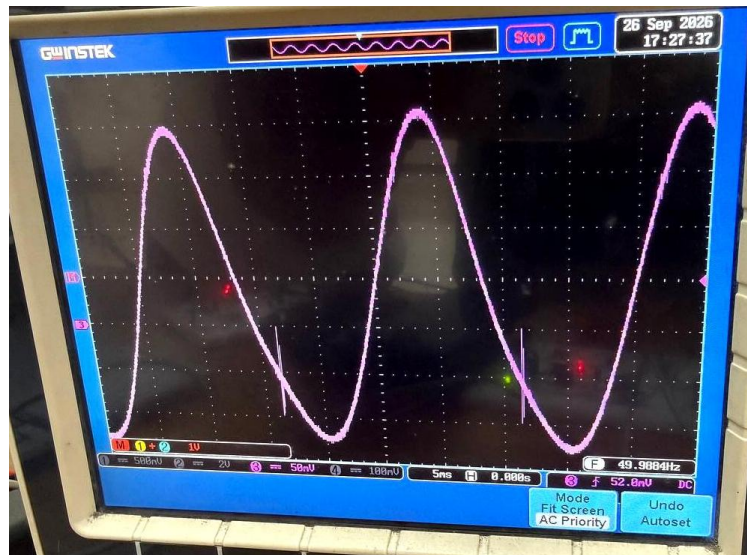


Figure 21: Recovered output at 50 Hz.

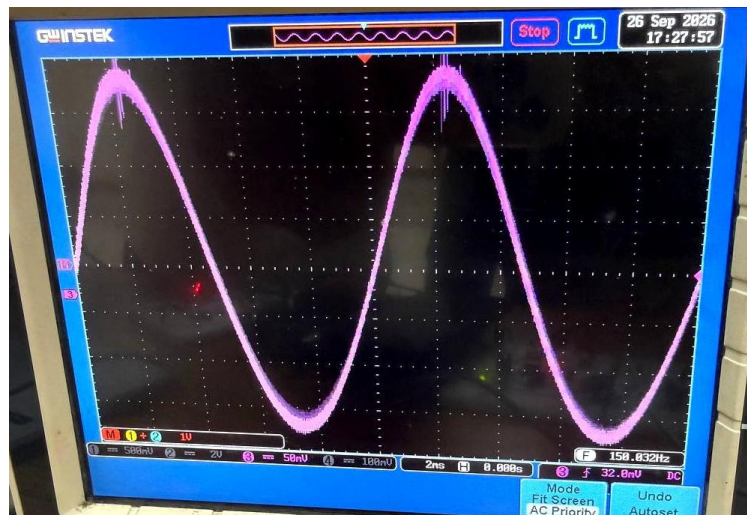


Figure 22: Recovered output at 150 Hz.

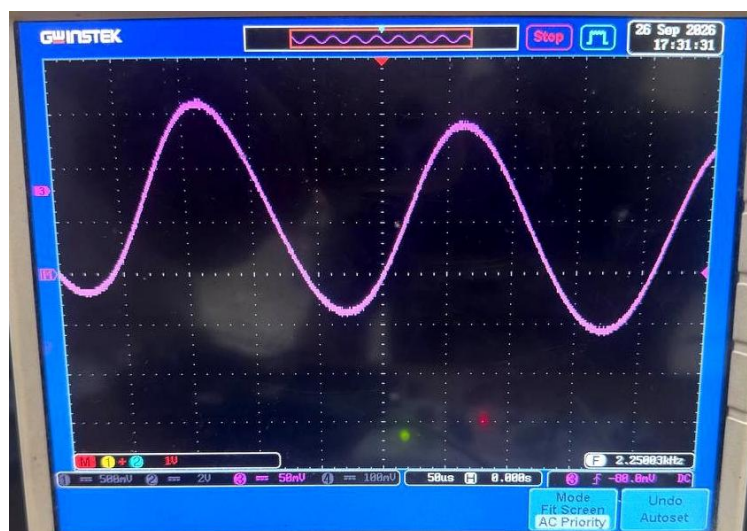


Figure 23: Recovered output at 2.25 kHz, within the flat mid-band.

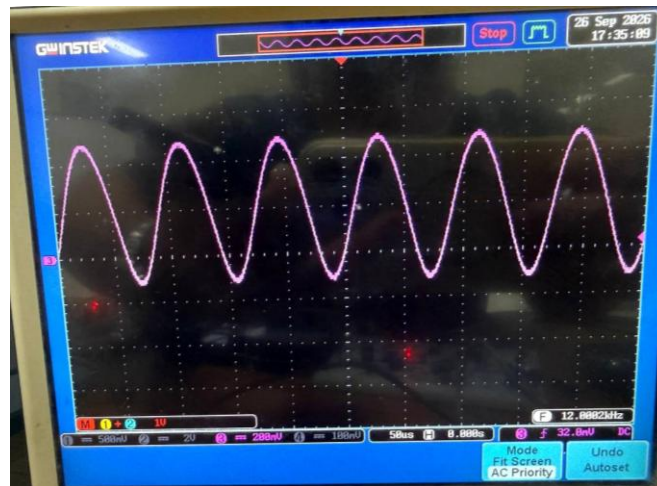


Figure 24: Recovered output at 12 kHz — the upper band edge.

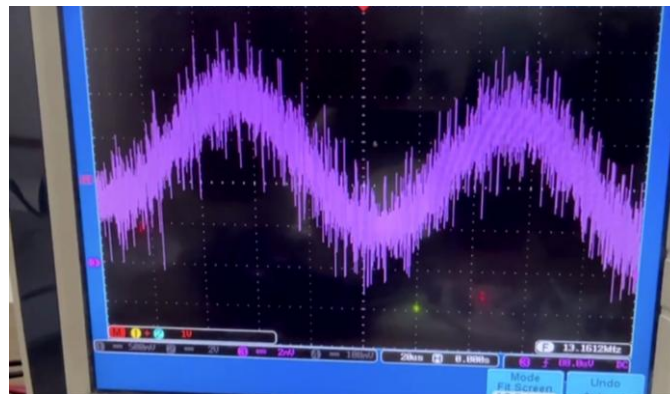


Figure 25: Recovered output at 13 kHz, beyond the filter cut-off: the amplitude has fallen and noise dominates.

The effect of ambient light was captured separately. Figure 26 shows the recovered waveform with a lamp switched on beside the receiver: a low-frequency ripple and visible noise are superimposed on the audio, and the sound was described by all group members as hazy with an audible hum. With the room light dimmed, the same waveform is clean.

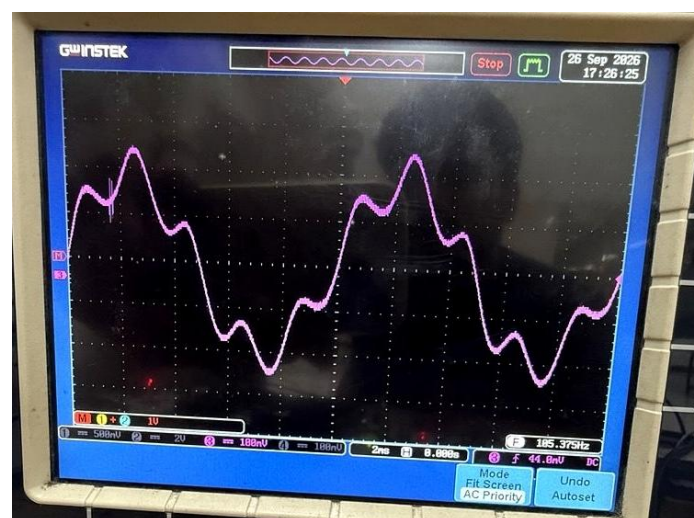


Figure 26: Recovered waveform with room lighting switched on close to the receiver; note the added ripple and noise.

Table 3 records the detector comparison.

Detector arrangement	Observed behaviour	Outcome
L14F1 phototransistor	Strong output but dull, muffled sound; slow response	Rejected — insufficient bandwidth
BPW34 photodiode, unshielded	Clearer treble, but sensitive to room lighting	Improved
BPW34 photodiode with black shroud	Clean audio, much less affected by ambient light	Adopted for the final build

Table 3: Comparison of the three detector arrangements tested.



Figure 27: First detector tried: response was not satisfactory.



Figure 28: Photodiode fitted with a black cover: measurably better, but not yet the best.



Figure 29: Final detector arrangement adopted for the completed link.

5.3.3 Results and Analysis

Table 4 compares the principal simulated and measured quantities.

Quantity	Simulated	Measured	Comment
Mid-rail reference	4.500 V	≈ 4.4 V	Agrees
Driver input bias (U2 pin 3)	1.154 V	≈ 1.15 V	Agrees
Laser bias current	30.4 mA	7.75 mA	Large deviation — discussed below
Upper -3 dB frequency	≈ 10.7 kHz	≈ 12 kHz	Close agreement
Lower -3 dB frequency	≈ 154 Hz	≈ 40 Hz	Deviation — discussed below
Waveform fidelity at 1 kHz	No visible clipping	No visible clipping	Agrees

Table 4: Comparison of simulated and measured performance.

The frequency-response results agree well at the upper edge. The simulation places the -3 dB point at about 10.7 kHz, set by the receiver Sallen-Key filter, and the measurement places it at about 12 kHz. A deviation of roughly 12 % is entirely consistent with the tolerance of the capacitors used: the 2.2 nF and 1 nF ceramics are ± 10 % parts, and a combined deviation in that direction moves the corner by about the observed amount. Figure 25 confirms that by 13 kHz the recovered signal has collapsed into noise, which is the expected behaviour immediately beyond a second-order corner.

The lower band edge shows a larger discrepancy. The design predicts a corner near 154 Hz from the 2.2 k Ω and 470 nF network in the receiver gain stage, whereas the measurement indicates useful output down to about 40 Hz. Two effects account for this. First, the -3 dB criterion applied during the measurement was based on the visible amplitude on the oscilloscope, which is a coarse reading on a photographed screen; second, the actual capacitor is a nominal-value part whose measured capacitance may be substantially larger. Both push the apparent corner downwards. For the purpose of the link, the relevant conclusion is that the response is adequate over the speech band in both the simulation and the hardware.

The laser bias current shows the largest deviation, and it is the most instructive result in the project. The simulation predicts 30.4 mA, the measurement gives 7.75 mA, a factor of four lower. The explanation lies in the difference between the ideal current-source model used in simulation and the real optical module. In the simulation the laser is modelled as a fixed 2.5 V source in series with 20 Ω , so the collector of Q1 can always source the current the feedback loop demands. In hardware the module presented a forward voltage of only about 2.06 V and, together with the series resistance actually present, the loop settled at a lower operating point than the ideal analysis predicts. The consequence is a reduced optical power and a reduced modulation depth, which in turn explains the modest signal-to-noise ratio observed at the receiver: the link works, and works over 10 m, but with less margin than the design intended.

The ambient-light experiment confirms the role of the high-pass corner in the receiver. With the lamp on, the disturbance appears as a low-frequency ripple rather than as a broadband increase in noise, which is precisely what the 100 Hz flicker of a mains-powered lamp would produce after partial attenuation by the 154 Hz corner. Had no high-pass action been present, the same illumination would have saturated the first amplifier stage entirely.

5.3.4 Interpretation and Conclusions on Data

The measurements support four conclusions.

- The link functions as designed. Speech and music are transmitted over a free-space laser beam and recovered intelligibly at a distance of 10 m indoors, with a usable range estimated at about 20 m in a darkened room.

- The usable audio bandwidth of approximately 40 Hz to 12 kHz covers the speech band completely and most of the music band, and is set deliberately by the receiver filter rather than accidentally by the parasitics of the devices.
- Constant-current drive is the decisive design choice. The first prototype, which drove the laser from a voltage source, produced audibly distorted sound; replacing it with the feedback-controlled current driver removed that distortion, and the recovered waveform in Figure 19 shows no visible asymmetry between the positive and negative half-cycles.
- Ambient light, not electronic noise, is the limiting factor on link quality in a normally lit room. Shielding the detector and filtering below 154 Hz together reduce this interference to an acceptable level, but they do not remove it; an optical band-pass filter over the detector would be the next step.

The chief limitation identified by the measurements is the low laser bias current and the consequent modest optical power. Re-establishing the intended 30 mA operating point — by lowering R11 further or by supplying the driver from a slightly higher rail — would raise the received signal directly and improve the signal-to-noise ratio without any change to the rest of the circuit.

5.4 Impact Assessment (PO(f))

5.4.1 Assessment of Societal and Cultural Issues

A light-based audio link has a positive societal implication in that it offers communication without adding to radio-frequency emissions, which is relevant in a densely populated city such as Dhaka where spectrum is heavily used. Because the link is confined to the room, it also supports privacy in a way that is easy for a non-technical user to understand: if the door is closed, the signal cannot leave. Culturally, the project is well suited to demonstration and teaching, since the operation of the link is visible — the audience can see the beam that carries the sound — which makes an abstract communication principle immediately intuitive. A negative aspect is that the technology requires an unobstructed line of sight, and any adoption of such a system must acknowledge that it cannot replace radio for mobile users.

5.4.2 Assessment of Health and Safety Issues

The principal health issue is optical exposure. A 5 mW laser is bright enough to cause retinal damage on direct viewing, and the risk is greatest for bystanders who are not aware of the beam. The design and the operating procedure both mitigate this: the beam is terminated at the detector, the optical axis is kept below eye level, the current driver prevents the module from being driven beyond its rating, and the user manual in Section 7.2 states these requirements explicitly. A secondary safety issue is the mains-connected bench supply; this was handled by using an isolating step-down transformer and by keeping the primary wiring away from the working area.

The circuit itself operates at 9 V and presents no electrical hazard. No thermal hazard was observed — no device on either board became warm to the touch during extended operation.

5.4.3 Assessment of Legal Issues

Unlike radio transmission, optical communication in the visible band requires no spectrum licence in Bangladesh or elsewhere, so the link raises no telecommunications regulatory issue. The relevant legal framework is instead product safety: laser products are classified internationally under IEC 60825-1, and a commercial version of this transmitter would require the corresponding Class 3R labelling, a protective housing, and a key or interlock. The prototype is a bench demonstrator operated under supervision and is not offered as a product. All components used are standard, off-the-shelf parts with no export or import restriction, and all reference designs consulted are published openly and are cited in Section 9.

5.5 Sustainability Evaluation (PO(g))

Resource use and reusability. The prototype uses through-hole components on reusable breadboards. No printed circuit board was etched, so no cupric chloride or ferric chloride waste was generated, and every component — including the two TL072 devices, the LM358, the LM386, the laser module and the photodiode — can be removed and reused in a later project. This is a material advantage over a soldered prototype at this stage of development.

Energy consumption. The measured receiver current is 10.48 mA at 7.96 V, about 83 mW, and the transmitter consumes a comparable amount, of which the laser diode accounts for the largest share. The complete link therefore operates on under a quarter of a watt. For comparison, a conventional radio audio link of similar range would typically consume a larger transmit power and require continuous protocol overhead. Because the transmitter contains no switching circuitry, it generates no electromagnetic interference.

Long-term sustainability. The design is purely analog, so there is no firmware to maintain, no software dependency to become obsolete, and no processor that will go out of production. The laser module is rated for over 2000 hours, and the remaining components are commodity parts that have been manufactured for decades and are likely to remain available. At end of life the boards can be disassembled by hand and the components recovered; the only items requiring controlled disposal are the electrolytic capacitors and the semiconductor devices. The principal sustainability limitation is the laser diode itself, which is a consumable with a finite lifetime and cannot be repaired when it fails.

6 Reflection on Individual and Team Work (PO(i))

6.1 Individual Contribution of Each Member

Student ID	Name	Individual contribution
2206119	Md. Ahasan Habib Kawsar	Circuit design, PSpice simulation, laser driver, integration, debugging, filter designing, demonstration (Project Leading) ,
2206111	Md. Mahedi Hasan Saikot	Data collection, oscilloscope measurements, frequency response testing, revision of slides
2206099	MD. Saruwer Hossan Pial	Receiver board assembly and soldering
2206104	Mustasin Rahman Ador	Transmitter board assembly, optical alignment and mounting
2206120	Md. Imran Shajid	Component procurement, BOM and cost analysis
2206121	Afnan Bin Obaid	Report writing, slides, demo video

Table 5: Individual contribution of each group member.

6.2 Mode of Team Work

The group worked through a combination of scheduled laboratory sessions and continuous online coordination. Design decisions were circulated to all members before being implemented, and every significant change to the circuit — the move to constant-current drive, the relocation of the dominant filter to the receiver, and the replacement of the input potentiometer by a fixed divider — was discussed by the group before the hardware was modified. Measurement sessions were attended together so that every member observed the behaviour of the link at first hand, and the results were recorded in a shared document that formed the basis of Section 5.3 of this report. Components were procured jointly, with the cost shared equally among the members.

6.3 Diversity Statement of Team

The group is drawn from different districts of Bangladesh and from different schooling backgrounds, and its members brought different technical strengths to the project: some were more comfortable with analog circuit theory and simulation, others with practical assembly, procurement and documentation. That range of strengths proved useful, because the problems encountered in this project were of correspondingly different kinds — a theoretical error in the filter values, a practical error in the wiring of the feedback path, and a procurement constraint when a 1.8 nF capacitor could not be obtained locally and a 2.2 nF part had to be substituted. Every member was given the opportunity to contribute to the decisions, and dissenting opinions about the design were resolved by measurement rather than by seniority.

6.4 Log Book of Project Implementation

Date	Milestone achieved	Comments
13 Jul	Project scope fixed; literature review; block diagram	Direct intensity modulation chosen over PWM
13 Jul	First PSpice model (UA741 transmitter, phototransistor receiver)	Simulation ran; design was voltage-driven
14 Jul	Components purchased from local market	Laser module, op-amps, passives
15 Jul	First breadboard prototype built	Audio transmitted over about 1 m
15 Jul	Debugging: output noisy and distorted	Traced to voltage-mode laser drive
16 Jul	Link confirmed working but quality unacceptable	Decision taken to redesign the driver
20 Jul	Redesign: TL072 preamp, Sallen-Key filter, LM358 current driver	Current drive linearises the laser
27 Jul	Filter cut-off found to be 34 kHz instead of 12 kHz	Capacitor values corrected
3 Aug	Staggered filters adopted (transmitter wide, receiver dominant)	Avoids the two filters stacking
10 Aug	Second hardware build on two separate boards	Separate supplies to break the ground loop
17 Aug	Distortion traced to excessive preamplifier gain	Gain reduced from 48 to 3.2
24 Aug	Laser bias investigated; R11 reduced from 56 Ω to 38 Ω	Brightness and modulation depth improved
31 Aug	Detector comparison: phototransistor, photodiode, shrouded photodiode	Shrouded BPW34 adopted
7 Sep	Input potentiometer replaced by fixed resistive divider	Removes sensitivity to source volume
14 Sep	Final integration and alignment of both units	Link stable over 10 m indoors
21 Sep	Final PSpice runs: bias point, AC sweep, transient	Simulated passband confirmed
26 Sep	Full measurement session on GW Instek GDS-2104A	40 Hz – 12 kHz measured; 10 m range verified
27 Sep	Report, presentation slides and demonstration video prepared	Video uploaded to YouTube
28 Sep	Final submission and hardware demonstration	

Table 6: Log book of project implementation.

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

Students at BUET have built a working communication link that sends sound on a beam of light. Music played on a mobile phone is converted into rapid variations in the brightness of a small red laser; across the room, a light sensor picks up those variations and a loudspeaker turns them back into sound. The link carries speech and music clearly over about ten metres, uses no radio waves at all, and costs under fifteen hundred taka to build from parts available in any local electronics market. Because light does not pass through walls, the signal stays in the room it was sent in — a natural form of privacy. Systems of this kind, known as Li-Fi, are of interest wherever radio is congested or restricted, such as hospitals, aircraft and industrial plants.

7.2 User Manual

Step 1 — Power. Connect a 9 V supply to each board and switch both on. The indicator LED on each board should light. The two boards must use separate supplies; do not connect a common ground wire between them.

Step 2 — Check the laser. The laser module should emit a steady red spot. Never look directly into the beam or at its reflection from a shiny surface. Keep the beam below eye level at all times.

Step 3 — Alignment. Place the receiver so that the laser spot falls at the centre of the photodiode. Slide the receiver slightly in each direction until the sound is loudest; this position is the correct alignment. Alignment matters more than distance.

Step 4 — Connect the audio source. Plug a mobile phone or laptop into the 3.5 mm socket on the transmitter using an audio cable, or use the on-board microphone for live speech. Set the source volume to about one third of its maximum.

Step 5 — Play. Start the audio. Adjust the volume control on the receiver for a comfortable listening level. If the sound is harsh or crackling, reduce the volume on the source, not on the receiver.

Step 6 — Best results. Dim the room lights if possible, and keep the photodiode shrouded from direct lamp light. Do not let anyone walk through the beam during a demonstration — the sound will cut out while the beam is blocked, which is itself a useful thing to show an audience.

Troubleshooting. No sound at all: check the alignment first, then confirm the audio source is playing and that both supplies are on. Hum or hiss: switch off nearby lamps and re-shroud the detector. Distorted sound: lower the source volume.

7.3 GitHub Link

The project is implemented entirely in hardware and contains no firmware or source code, so no software repository is associated with it. The PSpice schematic file and the measurement data are available from the group on request.

7.4 YouTube Link

Demonstration video:

<https://youtu.be/T3q3XpJ2jnU>

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Table 7 lists the components used in the two completed units and in the bench power supply. Table 8 lists the additional components purchased during development but not present in the final design; these are reported separately so that the cost of the delivered prototype is not overstated.

Component	Qty	Unit price (BDT)	Total (BDT)
TL072 dual op-amp	2	25	50
LM358 dual op-amp	1	15	15
LM386 audio power amplifier	1	20	20
BD135 NPN transistor	1	15	15
BPW34 photodiode	1	50	50
Laser diode module, 650 nm 5 mW	1	40	40
Resistors (assorted values)	30	1	30
Ceramic / film capacitors (assorted)	20	2	40
Electrolytic capacitors (10–1000 μ F)	13	5	65
Potentiometer, 10 k Ω	1	20	20
3.5 mm audio socket	1	15	15
3.5 mm audio cable	1	50	50
Loudspeaker, 8 Ω 0.5 W	1	60	60
Electret microphone	1	20	20
Breadboard	2	120	240
Jumper wire (one box)	1	150	150
Step-down transformer, 12 V 1 A	1	150	150
LM317 adjustable regulator	1	25	25
7809 fixed regulator	1	15	15
1N4007 rectifier diode	5	3	15
Indicator LED	2	3	6
9 V battery with snap connector	2	40	80
Total			1171

Table 7: Bill of materials for the delivered prototype.

Component	Qty	Unit price (BDT)	Total (BDT)
UA741 op-amp (first prototype)	1	15	15
BC548 / BC549 transistors (first prototype)	3	5	15
L14F1 phototransistor (detector trial)	1	20	20
LM386 (spare)	1	20	20
Additional capacitors of superseded values	20	2	40
Additional loudspeaker	1	60	60
Total additional development cost			170

Table 8: Additional components purchased during development and superseded in the final design.

8.2 Calculation of Per Unit Cost of Prototype

The delivered prototype consists of the transmitter unit, the receiver unit and the bench power supply. Taking the bill of materials in Table 7, the direct component cost is BDT 1171. Adding the development components in Table 8, which were genuinely consumed in reaching the final design, the total project expenditure is BDT 1341.

Cost element	Amount (BDT)
Components in the delivered prototype	1171
Components consumed during development	170
Total expenditure on the project	1341
Cost of one prototype unit (transmitter + receiver + supply)	1171

Table 9: Per unit cost of the prototype.

No labour cost is charged, since the work was carried out by the group as part of the course; if student time were costed at a nominal BDT 100 per hour for the approximately 40 hours spent on design, assembly and testing, an additional BDT 4000 would be attributable to labour.

8.3 Calculation of Per Unit Cost of Mass-Produced Unit

In volume production the cost structure changes substantially. The two breadboards (BDT 240) and the jumper wire (BDT 150) are replaced by a single-sided printed circuit board; component prices fall with bulk purchase; and the bench supply is replaced by a small commercial adaptor. The estimate below assumes a production run of 1000 units.

Cost element	Amount per unit (BDT)
Semiconductors and passives at bulk prices ($\approx 45\%$ discount)	210
Printed circuit board (two boards, panelised)	70
Enclosure, optics mount and connectors	120
9 V adaptor in place of the bench supply	90
Assembly and soldering (automated, amortised)	80
Testing and alignment	40
Packaging and documentation	30
Estimated factory cost per unit	640
With 25 % margin, indicative selling price	800

Table 10: Estimated per unit cost of a mass-produced unit at a volume of 1000.

The mass-produced unit is therefore estimated at approximately BDT 640, a reduction of about 45 % relative to the prototype. The saving comes almost entirely from eliminating the breadboards and the individually purchased passive components; the laser module and the photodiode, being finished optical parts, dominate the remaining bill and do not fall in price as steeply.

8.4 Timeline of Project Implementation

Figure 30 shows the timeline of the project from the fixing of the scope on 13 July 2026 to the final submission on 28 September 2026. The single longest activity was the redesign of the transmitter following the failure of the first prototype, which occupied the second half of July and the first week of August.

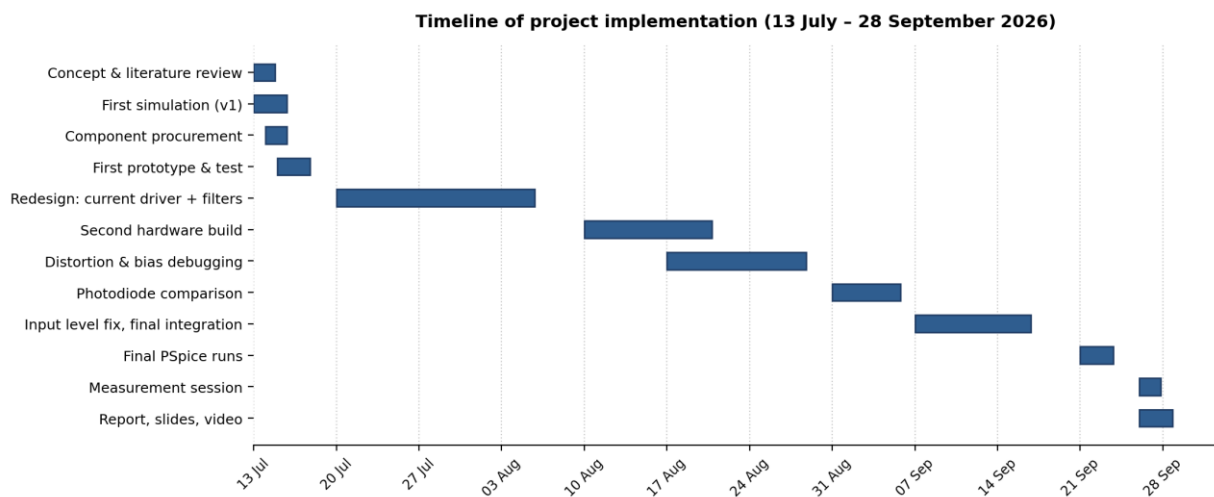


Figure 30: Gantt chart of project implementation, July to September 2026.

9 References

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- [10] International Electrotechnical Commission, *IEC 60825-1: Safety of Laser Products — Part 1: Equipment Classification and Requirements*, 3rd ed., 2014.
- [11] Department of EEE, BUET, "EEE 310 — Communication System I Laboratory: Course Outcomes and PO Mapping," <https://eee.buet.ac.bd/academics/undergraduate/courses/eee310>

Appendix A Mapping of Course Outcomes and Program Outcomes

CO	PO	Description	Addressed in
CO1	PO(a)	Explain practical aspects of communication schemes using modules and equipment	Section 3.1
CO2	PO(d)	Compare theoretical and experimental results	Section 5.3
CO3	PO(c)	Design with safety, cultural, societal and environmental considerations	Section 5.2
CO4	PO(f)	Assess societal, health, safety, legal and cultural impact	Section 5.4
CO5	PO(g)	Evaluate sustainability and environmental impact	Section 5.5
CO6	PO(i)	Demonstrate participation and leadership in teamwork	Section 6
CO7	PO(j)	Communicate effectively through presentation and report	Section 7
CO8	PO(k)	Demonstrate project management and cost analysis	Section 8

Table 11: Mapping of course outcomes to program outcomes and to the sections of this report.